Rational and Irrational Numbers

Focus: perfect-square roots, non-perfect roots, decimals and fractions, comparing and ordering

The test you are using. A number is **rational** when it can be written as $\frac{p}{q}$ with p and q whole numbers and $q \neq 0$. Every terminating decimal and every repeating decimal passes that test. A number is **irrational** when no such fraction exists — its decimal never ends and never repeats.

Model to work from. Keep this table of perfect squares beside you. A square root is exact — and therefore rational — exactly when the number under the radical appears in the bottom row.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
n^2	1	4	9	16	25	36	49	64	81	100	121	144	169	196	225

For every problem, state **rational** or **irrational** and give the reason in the work space. Write the value the problem asks for on the answer line.

1. Is $\sqrt{64}$ rational or irrational? Support your answer by writing $\sqrt{64}$ as a ratio of two whole numbers, then give its value.

Answer: ________________

2. Complete this row of the conversion table for the decimal 0.24, then classify it.

Decimal	**Fraction as read aloud**	**Lowest terms**	**Rational?**
0.24			

Write the lowest-terms fraction on the answer line.

Answer: ________________

3. The number 20 is not in the table of perfect squares. **(a)** Between which two consecutive whole numbers does $\sqrt{20}$ lie? **(b)** Is $\sqrt{20}$ rational or irrational? **(c)** Give $\sqrt{20}$ rounded to two decimal places, and explain why that rounded value is not the same number as $\sqrt{20}$. Write the rounded value on the answer line.

Answer: ________________

4. Look up $\sqrt{225}$ in the table above. Is it rational or irrational, and what is its exact value?

Answer: ________________

5. Order these three numbers from least to greatest, and mark the one that is irrational:

$$\sqrt{11}, \qquad 3.1, \qquad \frac{16}{5}$$

Show how you compared them — rewriting each as a decimal is a good start.

Answer: ________________

6. The decimal $0.\overline{63}$ repeats forever. **(a)** Explain why a decimal that repeats forever can still be rational. **(b)** It is known that $0.\overline{63} = \frac{63}{99}$. Write that fraction in lowest terms on the answer line.

Answer: ________________

7. **(a)** Between which two consecutive whole numbers does $\sqrt{40}$ lie? Justify it with two entries from the square table. **(b)** Is $\sqrt{40}$ rational or irrational? **(c)** Give $\sqrt{40}$ rounded to two decimal places on the answer line.

Answer: ________________

8. A classmate writes "$\sqrt{3} = 1.73$". Decide whether that statement is true, and write $>$, $<$ or $=$ in the blank to make a true statement:

$$\sqrt{3} \_\_\_\_\_\_ 1.73$$

Explain in one sentence what 1.73 really is.

Answer: ________________

9. Is $\sqrt{\dfrac{25}{36}}$ rational or irrational? Explain how the perfect-square table decides it for both the numerator and the denominator, then write the exact value on the answer line.

Answer: ________________

10. Neither $\sqrt{12}$ nor $\sqrt{3}$ is rational. **(a)** Compute $\sqrt{12} \cdot \sqrt{3}$ exactly and write the value on the answer line. **(b)** What does your answer show about this claim: *"the product of two irrational numbers is always irrational"*? Answer in one sentence.

Answer: ________________